

The short-time Wigner–Ville distribution

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ABSTRACT

The Wigner–Ville distribution (WD) and ambiguity function (AF) are serviceable tools for the global analysis of non-stationary signals. However, when dealing with time-varying signals that exhibit changing characteristics and require real-time signal processing, they have high limitations due to the existence of cross-items. In this paper, we propose the short-time Wigner–Ville distribution (STWD) as a novel approach that effectively analyzes time-varying signals with changing characteristics. First, the definition of STWD and short-time ambiguity function (STAF) and their properties are advanced. Next, three uncertainty principles that define the lower bound for the product of signal spread and its bandwidth are derived. Then, the discrete STWD (DSTWD) and the process of computerization are also proposed which can be used more widely in practice. Finally, the optimization of window functions based on Rényi entropy is discussed. Simulation experiments show that the STWD can be effectively applied for the analysis of local change characteristics and real-time processing of signals with valid and accurate results.

1. Introduction

The Wigner–Ville distribution (WD) and ambiguity function (AF) [1, 2] are powerful tools in time–frequency analysis, non-stationary signal processing, and some actual industrial scenarios [3,4]. Let the tensor product of f, g be

$$(f \otimes g^*)(t) = f(t + \frac{\tau}{2})f^*(t - \frac{\tau}{2}). \quad (1)$$

The essence of the WD is to perform the Fourier transform (FT) on the instantaneous autocorrelation function (IAF) with respect to time delay, while the AF is to perform the inverse Fourier transform (IFT) on IAF with concerning time. And they are defined as follows in terms of two given functions f and g of one variable:

$$W_{f,g}(t, u) = \int_{\mathbb{R}} (f \otimes g^*)(t) e^{-ju\tau} d\tau \quad (2)$$

$$AF_{f,g}(\tau, \theta) = \int_{\mathbb{R}} (f \otimes g^*)(t) e^{j\theta t} dt, \quad (3)$$

where $*$ denotes complex conjugate.

The WD effectuates energy accumulation in the time–frequency domain, and the AF indicates autocorrelation in the time-delay and frequency-shift domain [1,5]. However, the WD and the AF are global signal feature analysis methods, and they suffer from cross-terms. A prevalent cross-term arises when the signal comprises multiple frequency components coexisting within a shared time interval (Fig. 1 (a)).

Moreover, in scenarios where the frequency characteristics of the signal undergo substantial variations at different time instances, cross-terms may also manifest (Fig. 1(b)). Numerous research findings have been published regarding the topic of addressing cross-terms.

The pseudo WD (PWD) achieves this objective by using a time–frequency smoothing operation, typically a window function [1,2].

$$PW_{f,g}(t, u) = \int_{\mathbb{R}} (f \otimes g^*)(t) \phi^*(\tau) e^{-ju\tau} d\tau, \quad (4)$$

where $\phi(\tau)$ is a window function. But it still demonstrates interference, particularly when dealing with time-varying signals undergoing feature transformations. To further reduce the cross-terms and improve the time–frequency representation, the smoothed-pseudo WD (SPWD) applies a smoothing operation to the PWD [2,6].

$$SPW_{f,g}(t, u) = \iint_{\mathbb{R}^2} PW_{f,g}(\tau, \omega) h(t - \tau, u - \omega) d\tau d\omega, \quad (5)$$

where $h(t - \tau, u - \omega)$ is another window function. The SPWD smooths the generated cross terms to more effectively suppress cross-terms generated by multiple factors. For time-varying signals whose characteristics change with time, or need real-time and online processing, these classical methods cannot effectively analyze the signal since they cannot fully effectively deal with cross-terms caused by multiple factors.

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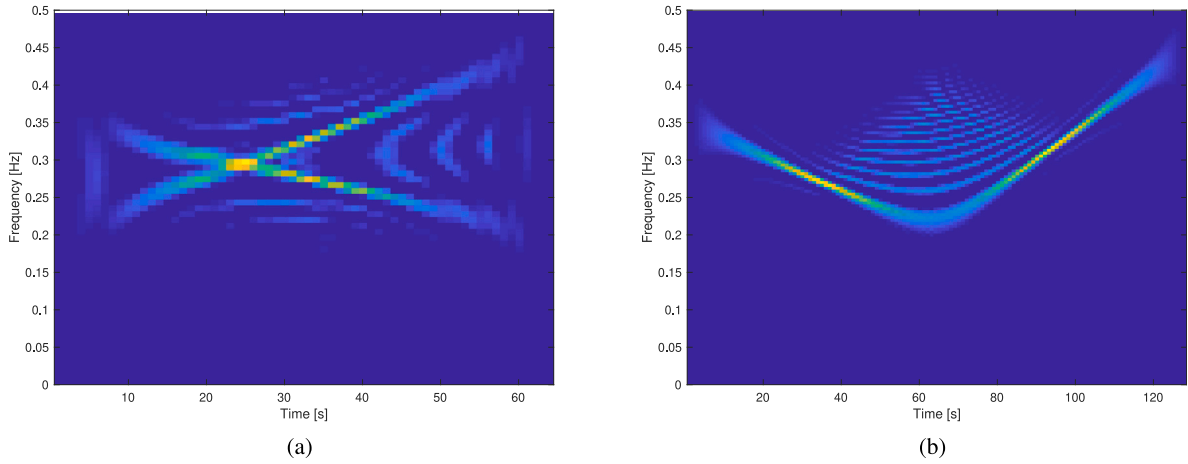


Fig. 1. (a) The WD of a signal contains multiple frequency components within the same time period. (b) The WD of a signal whose frequency characteristics undergo significant changes at different times.

In addition to the aforementioned methods, a novel approach (S-method) for suppressing cross terms based on the short-time Fourier transform (STFT) is proposed from a different perspective. The STFT represents the signal characteristics at a certain time through a segment of signal in the time window. The STFT of signal f is defined by [7–9]

$$STFT_f(t, u) = \int_{\mathbb{R}} f(\tau) \phi^*(\tau - t) e^{-ju\tau} d\tau, \quad (6)$$

where the window function $\phi(\tau)$ should be a symmetric real function. The STFT can analyze time-varying signals with changing characteristics. However, it suffers from a relatively low time–frequency resolution, limiting its ability to distinguish between multiple maneuvering targets that are closely spaced in the frequency domain.

Considering that the WD offers good time–frequency focusing and the STFT exhibits no cross-term interference, combining these two methods to develop a time–frequency analysis technique that leverages their respective strengths is a viable approach to enhance the performance of time–frequency analysis. Building upon this idea, L. Stanković proposed the S-method [10,11].

$$SM_f(t, u) = \frac{1}{\pi} \int_{\mathbb{R}} \Phi(2\theta) STFT_f(t, u + \theta) STFT_f^*(t, u - \theta) d\theta, \quad (7)$$

where $\Phi(\theta)$ is a narrow window in frequency domain. The S-method is defined based on the STFT of the signal, utilizing the relationship between the WD and the STFT. From an alternative perspective, this paper introduces a novel method that leverages the fundamental principle of the STFT to harness the advantages of both STFT and WD.

Based on the idea of the STFT, we can window the functions in WD in the time domain. Therefore, the short-time Wigner–Ville distribution (STWD) and short-time ambiguity function (STAF) are proposed in this paper. They can effectively analyze time-varying signals with changing characteristics, and can process the signals in real-time or online. And according to the relationship between the STWD and the STAF, we mainly discuss the properties of the STWD. In addition, we give three uncertainty principles, which are Heisenberg’s uncertainty principle, logarithmic uncertainty principle, and Lieb’s uncertainty principle [12–15]. These uncertainty principles give us information about a signal and its STWD; it declares that if we try to limit the behavior of one, the other will lose control. Its main sense in physics and especially in optics is that it defines the lower bound for the product of signal spread and its bandwidth [16,17].

Furthermore, the discrete method of STWD (DSTWD) and the process of computerization are also proposed which can be used more widely in practice. The adjustability of the window function makes the STWD more flexible, so we discuss the effect of different basic window functions and different window parameters. To better amplify

the advantages of the STWD, we further study a window function optimization method based on the above and Rényi entropy [18,19]. According to the nature of the different positions of the auto-term and the cross-term in the fuzzy domain, we can build a model to optimize the window parameters based on the STAF [20–22].

In this paper, the STWD and STAF are proposed and consolidated by the following steps. In Section 2, the definition of the STWD and STAF and their properties are summarized. Section 3 presents three uncertainty principles. In Section 4, the discrete STWD is derived. Section 5 discuss the effect of different basic window functions, and the adaptive STWD is proposed based on Rényi entropy. Finally, Section 6 concludes this paper.

2. The proposed short-time Wigner–Ville distribution

2.1. Definition of the STWD

The STWD and STAF were obtained by applying a window function to the instantaneous autocorrelation function (IAF). Motivated by this observation, we define the STWD and STAF as follows.

Definition 1. The cross STWD of signals $f, g \in L^2(\mathbb{R})$ is defined by

$$W_{f,g,\phi}(t, u) = \int_{\mathbb{R}} (f_{\phi} \otimes g_{\phi}^*)(t) e^{-ju\tau} d\tau, \quad (8)$$

where $\phi(t)$ is window function, p is the center of ϕ , and

$$f_{\phi}(t) = f(t)\phi(t - p), \quad g_{\phi}(t) = g(t)\phi(t - p). \quad (9)$$

When $g = f$, the STWD of signal f are given by

$$W_{f,\phi}(t, u) = \int_{\mathbb{R}} (f_{\phi} \otimes f_{\phi}^*)(t) e^{-ju\tau} d\tau. \quad (10)$$

Definition 2. The cross STAF of signals $f, g \in L^2(\mathbb{R})$ and the STAF of f are given by

$$AF_{f,g,\phi}(\tau, \theta) = \int_{\mathbb{R}} (f_{\phi} \otimes g_{\phi}^*)(t) e^{j\theta t} dt, \quad (11)$$

$$AF_{f,\phi}(\tau, \theta) = \int_{\mathbb{R}} (f_{\phi} \otimes f_{\phi}^*)(t) e^{j\theta t} dt. \quad (12)$$

According to the relationship between the traditional WD and AF, the relationship between the STWD and STAF can be quickly obtained

$$AF_{f,g,\phi}(\tau, \theta) = \iint_{\mathbb{R}^2} W_{f,g,\phi}(t, u) e^{-j(u\tau - \theta t)} du dt. \quad (13)$$

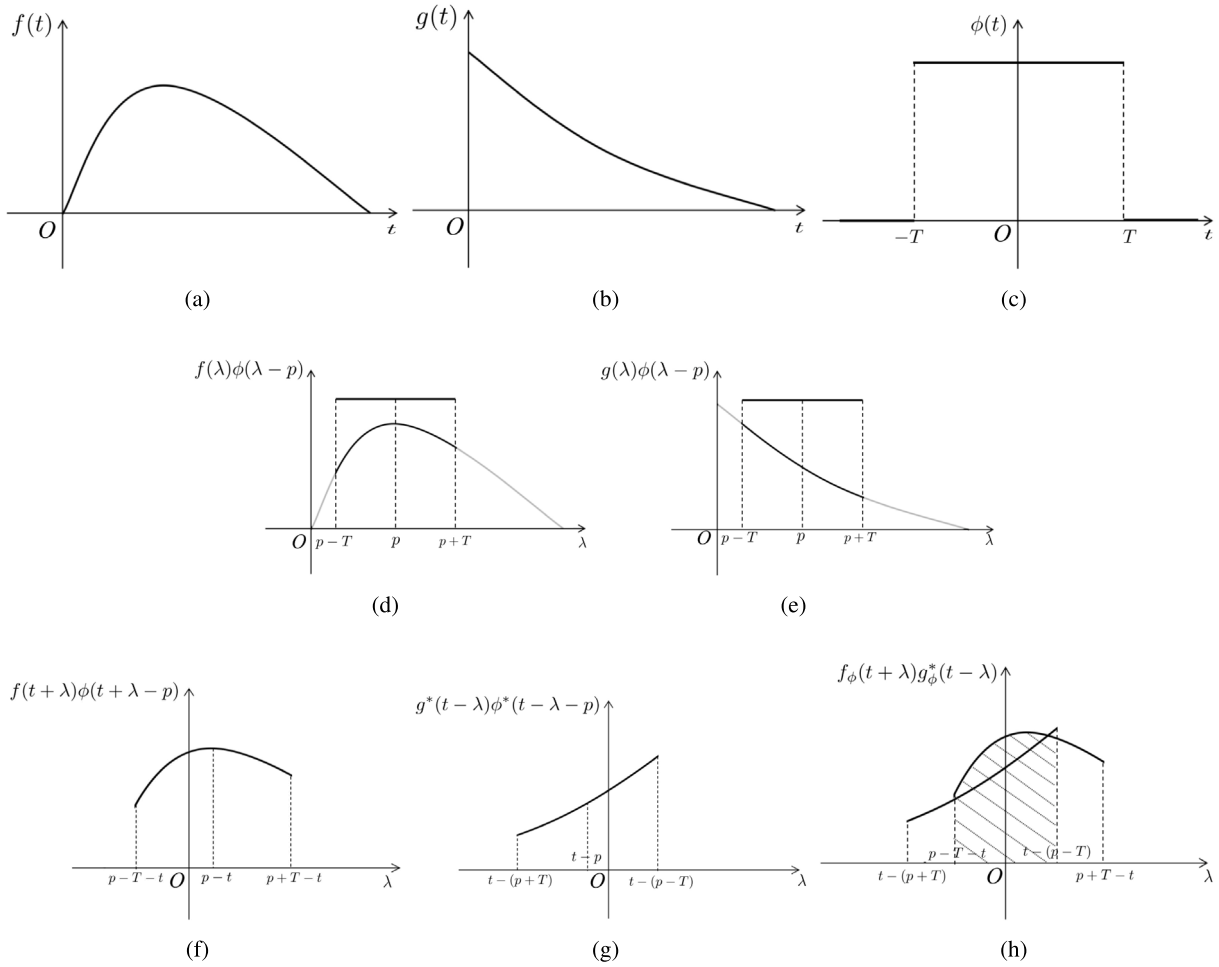


Fig. 2. Definition interpretation of STWD. (a) Signal $f(t)$. (b) Signal $g(t)$. (c) Window function $\phi(t)$. (d) Using the window function to intercept f . (e) Using the window function to intercept g . (f) Perform λ time shift on the intercepted f . (g) Perform λ time shift on the intercepted g . (h) Instantaneous autocorrelation.

That is, the STAF of f, g is the two-dimensional FT of the STWD. Therefore, we mainly deal with the STWD in the rest of this paper without loss of generality.

Next, let us explain the definition of (8). Setting $\tau/2 = \lambda$, we obtain

$$W_{f,g,\phi}(t, u) = \int_{\mathbb{R}} (f_{\phi} \otimes g_{\phi}^*)(t) e^{-j u \tau} d\tau. \quad (14)$$

Its meaning can be shown in Fig. 2. Apply FT to shaded portions of the figure, that is, the common part of the product of $f_{\phi}(t + \lambda)$ and $g_{\phi}^*(t + \lambda)$, to obtain the STWD at time p . Note that the simplest rectangular window is used in this example, and the kernel function of FT is $e^{-j2u\lambda}$.

2.2. Basic properties of the STWD

Several important properties of the STWD are discussed in this section. These properties play critical roles in signal representation.

Theorem 1 (Boundedness). Let $f, g, \phi \in L^2(\mathbb{R})$, then the $W_{f,g,\phi}(t, u)$ is bounded on $L^4(\mathbb{R})$, that is

$$|W_{f,g,\phi}(t, u)| \leq 16 \|f\|_2 \|g\|_2 \|\phi\|_2^2. \quad (15)$$

Theorem 2 (Marginal properties). Let $f, \phi \in L^2(\mathbb{R})$, the time marginal property of the STWD can be presented by

$$\frac{1}{2\pi} \int_{\mathbb{R}} W_{f,\phi}(t, u) du = |f(t)|^2, \text{ when } \phi(t) = 1. \quad (16)$$

And the frequency marginal property of the STWD can be expressed by

$$\int_{\mathbb{R}} W_{f,\phi}(t, u) du = |F(u)|^2, \text{ when } \phi(t) = 1. \quad (17)$$

Theorem 3 (Shift properties). Let $f, g, \phi \in L^2(\mathbb{R})$, then we have the shift properties as follows

(1) Consider the time shift operator $T_{\xi} f(t) = f(t - \xi)$, the time shift property is

$$W_{T_{\xi} f, T_{\xi} g, \phi}(t, u) = W_{f, g, \phi}(t - \xi, u). \quad (18)$$

(2) Consider the frequency shift operator $S_{\xi} f(t) = f(t) e^{j\xi t}$, the frequency shift property is

$$W_{S_{\xi} f, S_{\xi} g, \phi}(t, u) = W_{f, g, \phi}(t, u - \xi). \quad (19)$$

Theorem 4 (Nonlinear). Let $f(t) = f_1(t) + f_2(t)$, then

$$W_{f,\phi}(t, u) = W_{f_1,\phi}(t, u) + W_{f_2,\phi}(t, u) + W_{f_1,f_2,\phi}(t, u) + W_{f_2,f_1,\phi}(t, u). \quad (20)$$

Theorem 5 (Parseval's relation). Let $f, \phi \in L^2(\mathbb{R})$, then the Parseval's relation is expressed by

$$\|W_{f,\phi}\| = \|f\|_2^2 \|\phi\|_2^2. \quad (21)$$

3. Uncertainty principles for the STWD

3.1. Heisenberg's uncertainty principle

In this section, we give a short and simple proof of Heisenberg's uncertainty principle of the STWD. Heisenberg's uncertainty principles for the FT are essentially a family of inequalities on the product of the spread $\Delta_t^2 = 4\pi \int_{\mathbb{R}} \frac{(t-t^0)^2 |f(t)|^2}{\|f\|_2^2} dt$ in the time domain and the spread $\Delta_u^2 = 4\pi \int_{\mathbb{R}} \frac{(u-u^0)^2 |F(u)|^2}{\|F\|_2^2} du$, where $t^0 = \int_{\mathbb{R}} \frac{t|f(t)|^2}{\|f\|_2^2} dt$ and $u^0 = \int_{\mathbb{R}} \frac{u|F(u)|^2}{\|F\|_2^2} du$ [12]. Then there are the following inequality relations $\Delta_t^2 \Delta_u^2 \geq \frac{1}{4}$, that is

$$\begin{aligned} & \int_{\mathbb{R}} (t-t^0)^2 |f(t)|^2 dt \int_{\mathbb{R}} (u-u^0)^2 |F(u)|^2 du \geq \frac{1}{16\pi^2} \|f\|_2^4 \\ & = \frac{1}{16\pi^2} \left(\int_{\mathbb{R}} |f(t)|^2 dt \right)^2. \end{aligned} \quad (22)$$

Let t_ϕ^0 denote the moments in time-STWD domain. For

$$\begin{aligned} & \int_{\mathbb{R}^2} t |W_{f,\phi}(t, u)|^2 dt du \\ & = \int_{\mathbb{R}^2} t \left| f\left(t + \frac{\tau}{2}\right) \phi\left(t - p + \frac{\tau}{2}\right) f^*\left(t - \frac{\tau}{2}\right) \phi^*\left(t - p - \frac{\tau}{2}\right) \right|^2 dt d\tau, \end{aligned}$$

setting $x = t + \tau/2$, $y = t - \tau/2$, we have

$$\begin{aligned} & \int_{\mathbb{R}^2} t |W_{f,\phi}(t, u)|^2 dt du \\ & = \int_{\mathbb{R}^2} \left(\frac{1}{2}x + \frac{1}{2}y \right) |f(x)|^2 |\phi(x-p)|^2 |f(y)|^2 |\phi(y-p)|^2 dx dy. \end{aligned}$$

Then based the Parseval's relation in (21), we obtain

$$\begin{aligned} t_\phi^0 & = \iint_{\mathbb{R}^2} \frac{t |W_{f,\phi}(t, u)|^2}{\|W_{f,\phi}(t, u)\|_2^2} dt du = \iint_{\mathbb{R}^2} \frac{t |W_{f,\phi}(t, u)|^2}{\|f\|_2^4 \|\phi\|_2^4} dt du \\ & = \iint_{\mathbb{R}^2} \frac{\left(\frac{1}{2}x + \frac{1}{2}y \right) |f(x)|^2 |\phi(x-p)|^2 |f(y)|^2 |\phi(y-p)|^2}{\|f\|_2^4 \|\phi\|_2^4} dx dy \\ & = \frac{1}{2} t^0 + \frac{1}{2} t^0 = t^0. \end{aligned} \quad (23)$$

Considering this significant characteristic, we can now get the following crucial result which is derived in Appendix A.

Theorem 6 (Heisenberg's uncertainty principle for STWD). Let $f, \phi \in L^2(\mathbb{R})$ and $W_{f,\phi}(t, u) \in L^2(\mathbb{R})$ be the STWD of f . Then

$$\begin{aligned} & \left(\int_{\mathbb{R}} (t-t_\phi^0)^2 |f(t)|^2 dt \right)^{\frac{1}{2}} \left(\iint_{\mathbb{R}^2} (u-u_\phi^0)^2 |W_{f,\phi}(t, u)|^2 du dt \right)^{\frac{1}{2}} \\ & \geq \frac{1}{4\pi} \|f\|_2^3 \|\phi\|_2^2. \end{aligned} \quad (24)$$

3.2. Logarithmic uncertainty principle

The logarithmic uncertainty principles of the FT have been studied in [13,15]. In this section, we are committed to deriving the logarithmic uncertainty principle of the STWD. Let $f \in L^2(\mathbb{R})$ and $F(u) \in L^2(\mathbb{R})$ be the FT of f . Then

$$\int_{\mathbb{R}} |f(t)|^2 \ln |t| dt + \int_{\mathbb{R}} |F(u)|^2 \ln |u| du \geq D \int_{\mathbb{R}} |f(t)|^2 dt, \quad (25)$$

where $D = \psi(\frac{1}{4}) - \ln \pi$, $\psi(t) = \frac{d}{dt} \ln(\Gamma(t))$, and $\Gamma(t)$ is the Gamma function.

Then we focus on the following logarithmic uncertainty principle of the STWD, and the proof is presented in Appendix B.

Theorem 7 (Logarithmic uncertainty principle for STWD). Let $f, \phi \in L^2(\mathbb{R})$ and $W_{f,\phi}(t, u) \in L^2(\mathbb{R})$ be the STWD of f . Then the following

inequality is held

$$\|\phi\|^4 \|f\|^2 \int_{\mathbb{R}} \ln |f(t)|^2 dt + \iint_{\mathbb{R}^2} |W_{f,\phi}(t, u)|^2 \ln |u| du dt \geq D \|f\|^4 \|\phi\|^4, \quad (26)$$

where $D = \psi(\frac{1}{4}) - \ln \pi$, $\psi(t) = \frac{d}{dt} \ln(\Gamma(t))$, and $\Gamma(t)$ is the Gamma function.

3.3. Lieb's uncertainty principle

This section presents Lieb's uncertainty principle for the STWD. Let us first review the concept of ϵ -concentrated of a signal function on any measurable set $M \subseteq \mathbb{R}^2$.

Definition 3. A signal $f \in L^2(\mathbb{R}^2)$ is said to be ϵ -concentrated [14] on a measurable set $M \subseteq \mathbb{R}^2$, if

$$\left(\int_{\mathbb{R}^2 \setminus M} |f(t)|^2 dt \right)^{\frac{1}{2}} \leq \epsilon \|f\|_{L^2(\mathbb{R}^2; \mathbb{H})}. \quad (27)$$

First, we can get the weak uncertainty principle for the STWD as follows.

Theorem 8 (Weak uncertainty principle for STWD). Suppose that $\|f\|_2 = \|\phi\|_2 = 1$. If $M \subseteq \mathbb{R}^2$, $\epsilon > 0$ are such that (see Appendix C.1)

$$\iint_M |W_{f,\phi}(t, u)|^2 dt du \geq 1 - \epsilon, \quad (28)$$

then

$$|M| > 1 - \epsilon. \quad (29)$$

To derive Lieb's uncertainty principle of the STWD, we need to introduce the following lemmas [16].

Lemma 1. Let $2 \leq p \leq \infty$, and $1/q = 1 - 1/p = (p-1)/p$. If $f \in L^q$ then $f \in L^q$ and

$$\|F\|_p \leq C_q \|f\|_q, \quad (30)$$

where $F(u)$ is the FT of f . Moreover, Let $1 \leq p \leq 2$ and suppose that $f \in L^r$ for some $1 \leq r \leq 2$, in which case F exists by (30). If $F(u) \in L^p$ then $q \in L^q$ with $q = (p-1)/p$ and

$$\|F\|_p \geq C_q \|f\|_q. \quad (31)$$

Lemma 2. Let $1/m + 1/n = 1 + 1/r$ with $1 \leq m \leq \infty, \leq n \leq \infty, \leq r \leq \infty$. Then, when $f \in L^m$ and $g \in L^n$, $f * g \in L^r$ and

$$\|f * g\|_r \leq (C_m C_n / C_r) \|f\|_m \|g\|_n. \quad (32)$$

Now we arrive at the following important result (see Appendix C.2).

Theorem 9. (Lieb's uncertainty principle for STWD) Let $f, \phi \in L^2(\mathbb{R})$, $W_{f,\phi}(t, u) \in L^2(\mathbb{R})$ be the STWD of f and $2 \leq p < \infty$. Then

$$\iint_{\mathbb{R}^2} |W_{f,\phi}(t, u)|^2 \ln |u| du dt \geq \frac{2}{p} [\|f\|_2^2 \|\phi\|_2^2]^{p/2}. \quad (33)$$

4. Discrete short-time Wigner-Ville distribution

4.1. Discrete algorithm

In physical systems, the implementation of STWD relies on discrete numerical methods, both in the time domain as well as in the frequency domain.

First, we discuss the discrete numerical method in the time domain. Sampling signal $f(t)$ with sampling interval T_0 , that is $t = nT_0, n = 0, 1, 2, \dots, N-1$, where N is the number of sampling points. For the continuous STWD in (10), let $\tau/2 = mT_0$, S be the step of window

function movement on the time axis, and T be the length of the window function, then the number of windows divided is

$$n_\phi = \lfloor \frac{N-T-S}{S} \rfloor, \quad (34)$$

where $\lfloor \cdot \rfloor$ rounds a number to the nearest integer. Hence, let $q = 0, S, 2S, \dots, n_\phi S$, the integral of τ in (10) becomes the sum of m , that is

$$W_{f,\phi}(nT_0, \omega) = 2T_0 \sum_m f(nT_0 + mT_0) \phi(nT_0 - qT_0 + mT_0) f^*(nT_0 - mT_0) \\ \times \phi^*(nT_0 - qT_0 - mT_0) e^{-j2m\omega}, \quad (35)$$

where $\omega = uT_0$ is circular frequency. The step S determines the distance between window functions moving along the time axis. The smaller S is, the more values of q will be taken. It is worth noting that in the definition of continuous STWD, p represents the center of the window function, and here, for the sake of simplicity of form, q represents the starting point of the window function. And the following window functions all start from zero instead of centering on zero.

Next, we consider the frequency domain. Normalize T_0 to 1, and let $\omega_k = kT_\omega$, $T_\omega = 2\pi/M$, where M indicates that a period 2π in the frequency domain is divided into M points. Therefore, we get the definition of the discrete STWD (DSTWD) as follows.

Definition 4 (DSTWD). Consider $f(n)$, $n, m, k, l = 0, 1, 2, \dots, N-1$, then the discrete STWD of f is defined by

$$W_{f,\phi}(n, k) = 2 \sum_m (f_\phi \otimes f_\phi^*)(n) e^{-j2mkT_\omega}, \quad (36)$$

$$f_\phi(n) = f(n)\phi(n-q), \quad (37)$$

where $q = 0, S, 2S, \dots, n_\phi S$, S is the step of window function movement on the time axis, $n_\phi = \lfloor \frac{N-T-S}{S} \rfloor$ is the number of windows, T is the length of window function, and $T_\omega = 2\pi/M$, M indicates that a period 2π in the frequency domain is divided into M points. Similar to (36), the discrete STAF (DSTAF) is defined by

$$AF_{f,\phi}(m, l) = 2 \sum_n (f_\phi \otimes f_\phi^*)(n) e^{-j2nlT_\omega}. \quad (38)$$

Algorithm 1: The fast calculation of the DSTWD.

Input: The discrete sequence $f(n)$, window function $\phi(n-q)$, window length T and frameshift S ;

Output: The DSTWD of f ;

- 1: Process the signal sequence in frames. $N = \text{length}(f)$ and all frames can be calculated by window length T and frameshift S , that is, the number of windows $n_\phi = \lfloor \frac{N-T-S}{S} \rfloor$;
- 2: Multiply each frame of f by window function ϕ , and denoted as $f_{\phi,i}$, $i = 1, 2, \dots, n_\phi$;
- 3: Calculate the autocorrelation of each frame, that is $f_{\phi,i}(n+m)f_{\phi,i}^*(n-m)$;
- 4: Apply the FFT algorithm to the autocorrelation function above, we get $\text{DSTWD}(f) = \text{FFT}(f_{\phi,i}(n+m)f_{\phi,i}^*(n-m))$;
- 5: **return** $\text{DSTWD}(f)$.

Since the STWD is closely related to the FT, the time–frequency spectrogram obtained from STWD is related to the FT domain frequencies. Therefore, the fast Fourier transform (FFT) algorithm can be used to realize fast calculation of the DSTWD. See Algorithm 1 for the algorithm flow.

4.2. Simulation

In this section, we give an example to show the computer discrete results of the STWD and its advantages over the STFT and the classical WD. Consider a time-varying signal with varying characteristics

$$f(n) = \begin{cases} e^{-j\pi(n^2/420+7n/20)}, & 0 \leq n \leq 63 \\ e^{j\pi(n^2/630-257n/3150)}, & 64 \leq n \leq 127 \\ e^{-j\pi(n^2/420+55n/84)}, & 128 \leq n \leq 191 \\ e^{j\pi(n^2/315-1259n/2100)}, & 192 \leq n \leq 255 \end{cases}. \quad (39)$$

The STFT of the signal f can reveal the frequency change of the signal, but it is fuzzy and has cross-terms (Fig. 3(a)). And the cross-term of the WD on the signal f is very serious, and the signal cannot be analyzed effectively (Fig. 3(b)). The PWD is an alternative approach that suppress the cross-terms (Fig. 3(c)). Compared to the WD, the cross-term of the PWD has slightly decreased. However, it still lacks the ability to effectively handle cross-terms resulting from variations in signal characteristics across different time periods. Fig. 3(d) is the SPWD which applies a smoothing operation to the PWD on f . Although the cross-term problem is further suppressed, it leads to a reduction in time–frequency resolution. The S-method is a hybrid approach that leverages the advantages of both the STFT and the WD. However, it still exhibits some cross terms when there are variations in signal features, as shown in Fig. 3(e). Despite the effectiveness of the S-method, it is important to acknowledge its limitations in capturing accurate signal characteristics under certain conditions. In contrast, Fig. 3(f) demonstrates the use of the STWD on signal f with the Hanning window function. This representation clearly depicts the frequency changes without the interference of cross terms. In such cases, the STWD with appropriate window functions can provide a more reliable representation of frequency variations without the undesirable effects of cross terms. The use of appropriate window functions further enhances the performance of the STWD in capturing dynamic changes in signals. This comparison highlights the advantages of the STWD over the S-method in scenarios where accurate frequency analysis is crucial.

Furthermore, we extend the application of STWD to stepped-frequency linear frequency modulation (SF-LFM) signals to explore its potential applications. The SF-LFM signal employs LFM signals instead of simple rectangular sub-pulses or frequency-stepping signals, which holds significant importance in radar systems [23]. This approach not only preserves the advantages of the frequency stepping signal, such as low sampling rate and narrowband processing, but also incorporates the features of the LFM signal, including large time and bandwidth. Let the total bandwidth of SF-LFM be denoted as B , the number of narrowband LFM subpulses be N , the bandwidth of each subpulse be $B_N = B/N$, the width of each subpulse be T_p , and the frequency modulation slope be $K = B_N/T_p$. To ensure there are no gaps or overlaps in the synthesized broadband spectrum, the carrier frequency increment Δf between adjacent subpulses is typically set as B_N . Let the center frequency of the first subpulse be u_0 , then the center frequency of the $i+1$ subpulse is $u_{i+1} = u_0 + iB_N$, where $i = 0, 1, \dots, N-1$. Then the SF-LFM is defined by [24]

$$f(t) = \sum_{i=0}^{N-1} f_1(t - iT_r) \cdot \exp[j2\pi u_i t] \\ = \sum_{i=0}^{N-1} \text{rect}\left(\frac{t - iT_r}{T_p}\right) \cdot \exp[j\pi K(t - iT_r)^2] \cdot \exp[2\pi f_i t], \quad (40)$$

where $u_1(t) = \text{rect}\left(\frac{t}{T_p}\right) \cdot \exp[j\pi Kt^2]$ is baseband LFM subpulse, and T_r represents the pulse repetition period. Next, we construct an SF-LFM signal with parameters $B = 0.35$, $N = 7$, $u_0 = 0.025$, $T_p = 256$, and $T_r = 384$. We can observe the time–frequency spectrograms obtained using different methods (Fig. 4). Among them, both the S-method and STWD demonstrate remarkable performance, showcasing strong time–frequency clustering capabilities without any cross-term interference. These methods excel in capturing and representing the energy distribution of signals in the time–frequency domain, enabling better visualization and analysis of complex signal behaviors.

Remark 1. In this section, we present the simulation results of a segmented analog signal, highlighting the advantages of the method proposed in this paper. Furthermore, an actual SF-LFM signal is discussed and compared to the STFT, WD, PWD, and SPWD methods. Both the proposed method and the S-method in this article demonstrate excellent performance. However, when it comes to practical application

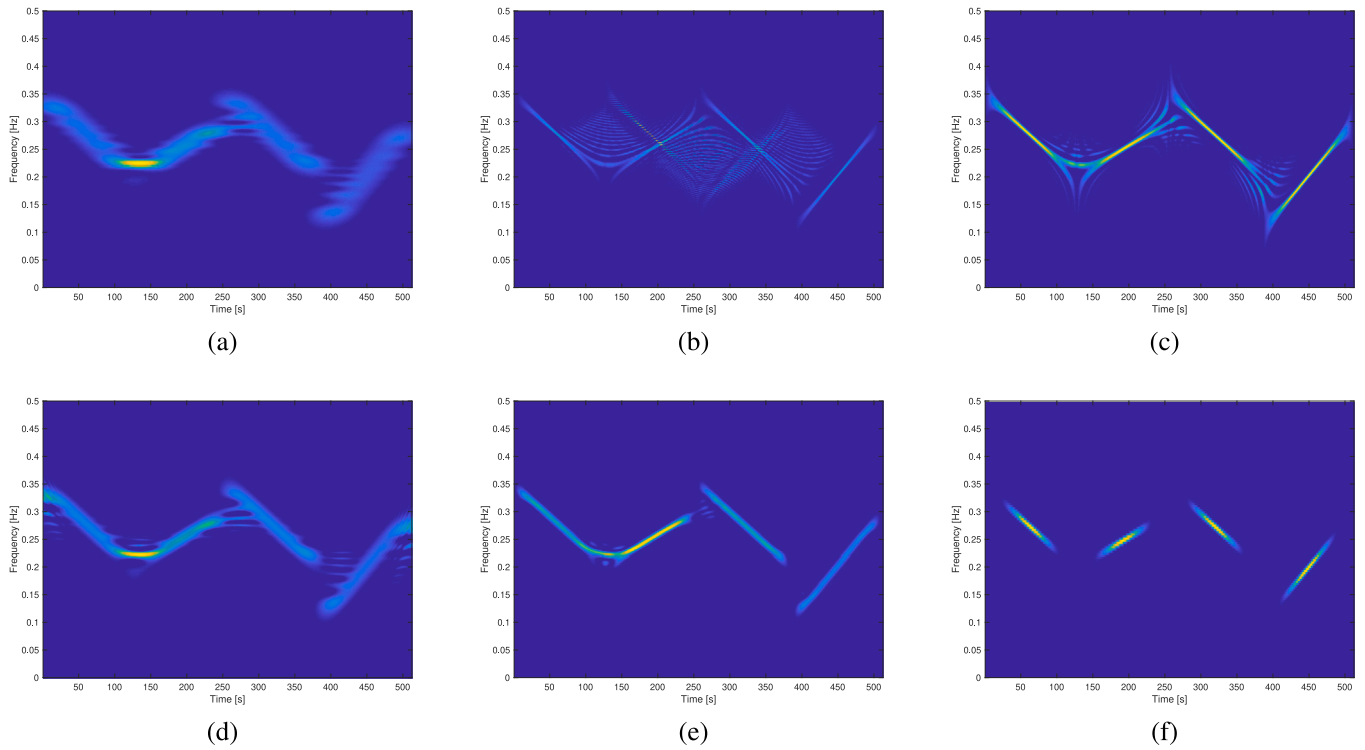


Fig. 3. (a) The STFT on the time-varying signal f with varying characteristics in (39). (b) The WD on f . (c) The PWD on f . (d) The SPWD on f . (e) The S-method on f . (f) The STWD on f .

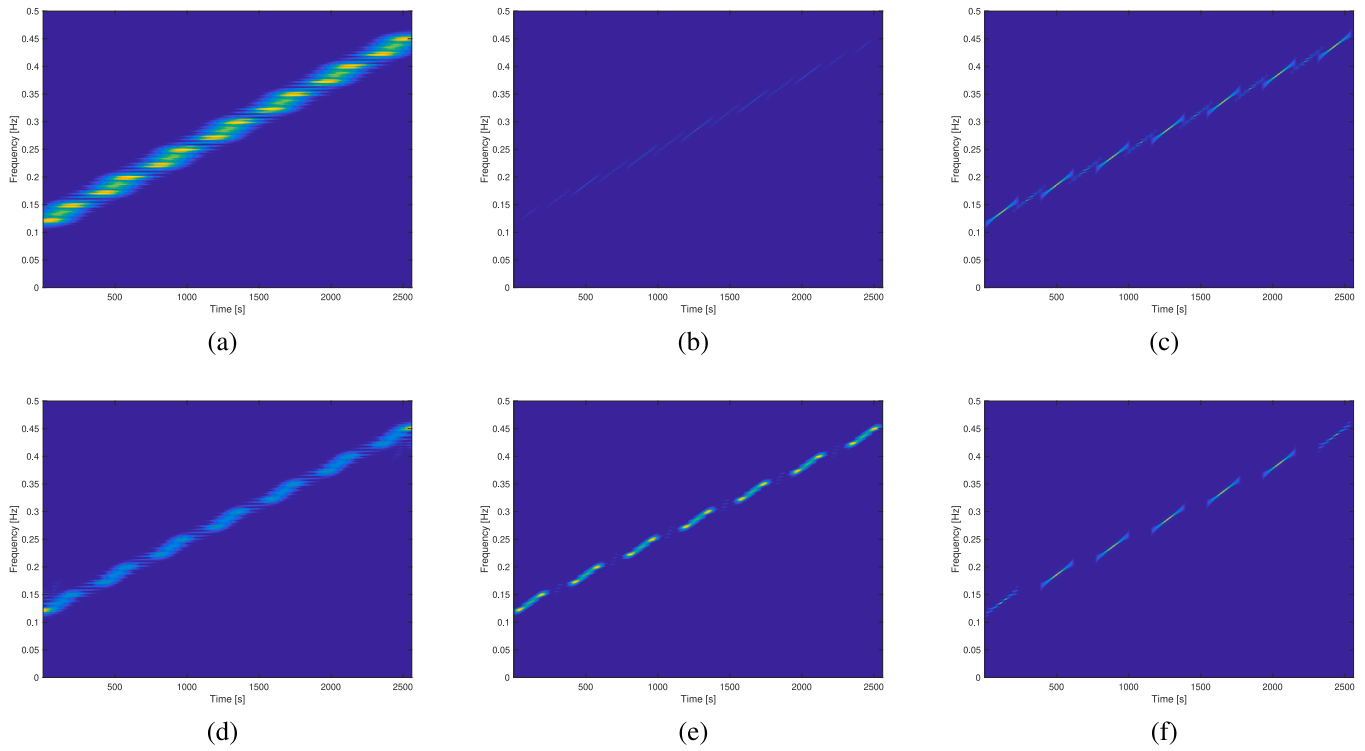


Fig. 4. (a) The STFT on the SF-LFM signal f with varying characteristics in (39). (b) The WD on f . (c) The PWD on f . (d) The SPWD on f . (e) The S-method on f . (f) The STWD on f .

scenarios, the complexity increases significantly. Therefore, the next section introduces an adaptive method that aims to address this challenge by enabling the adaptive selection of parameters. This adaptive approach will enhance the flexibility and effectiveness of the proposed method in real-world applications.

5. Adaptive short-time Wigner–Ville distribution

5.1. Basic window functions

In this section, we compare the effects of different window functions. The time domain diagram and frequency domain diagram of some basic window functions are shown in Table 1. If there is no special description in the table, n is from 0 to $N - 1$. Where I_0 in Kaiser window is the first kind of deformed zero-order Bessel function, and when $\alpha > 50$, $\beta = 0.1102(\alpha - 8.7)$, when $21 \leq \alpha \leq 50$, $\beta = 0.5482(\alpha - 21)^{0.4} + 0.07886(\alpha - 21)$, when $\alpha < 21$, $\beta = 0$. And in the Chebyshev window, $T_{N/2-1}(c_T) = \begin{cases} \cos[(N/2 - 1) \arccos c_T], & |c_T| \leq 1 \\ \cosh[(N/2 - 1) \operatorname{arccosh} c_T], & |c_T| > 1 \end{cases}$, and r_c is the ratio of side lobe amplitude to main lobe amplitude.

Then we carry out experiments with different window functions for the signal f in (39), and the results are shown in Fig. 5. The window length and frameshift are taken uniformly, that is $T = 64, S = 64$. It can be seen that the results of signal analysis by different window functions are quite different. Among them, the Hanning window, Hamming window and Gauss window have good effects, which can clearly show the frequency trend and change. While the rectangular window, triangular window and Kaiser window have some cross-term problems. Moreover, the Blackman window, Flat Top window and Chebyshev window have too concentrated frequency to effectively analyze.

In addition to the influence of different window functions on the results, the selection of window length T and frameshift S also has a great influence on the results. As shown in Fig. 6, we set six different combinations of window length T and frameshift S parameters. Both window length and frameshift affect frequency resolution. A shorter window function and frameshift can quickly track signal changes, but the frequency resolution is low; a longer window function and frameshift can obtain better frequency resolution, but blur the temporal variation of features.

5.2. Optimization of window function

5.2.1. Optimization method

Through the analysis in the previous section, the selection of different window functions, window length and frameshift will have an impact on the signal analysis effect. To effectively analyze signal characteristics and achieve real-time processing, the window function of the STWD must change adaptively with the different signal characteristics of a certain time slice. First, we select the Gauss window with good effect, that is, $\phi(n) = e^{-\frac{1}{2}(\frac{\delta n}{T})^2}$, $\delta \geq 2$. Then consider the window length T and frameshift S , and for simplicity, let $S = T$. That is, optimize window length to design appropriate window functions. Finally, an adaptive parameter optimization algorithm is designed.

Consider a time-varying signal with changing characteristics

$$f(n) = \begin{cases} f_1(n) = e^{j(\frac{v_1}{2}n^2 + r_1n)}, & 0 \leq n \leq N_1 \\ f_2(n) = e^{j(\frac{v_2}{2}n^2 + r_2n)}, & N_1 < n \leq N_1 + N_2. \end{cases} \quad (41)$$

The AF of f in a window T_1 can also be divided into auto term and cross term.

$$AF_f^{T_1}(m, l) = AF_{f_1, \phi}(m, l) + AF_{f_1, f_2, \phi}(m, l) + AF_{f_2, \phi}(m, l). \quad (42)$$

The auto-term of $AF_f^{T_1}(m, l)$ is

$$AF_f^{T_1}(m, l) = 4\pi e^{j2rm} \delta(2v_i m + 2lT_\omega), i = 1, 2, \quad (43)$$

and the cross-term of $AF_f^{T_1}(m, l)$ is

$$AF_{f_1, f_2}^{T_1}(m, l) = 2\sqrt{\frac{2\pi}{v_2 - v_1}} e^{j\frac{v_1 - v_2}{2}m^2} e^{j(r_1 + r_2)m} e^{-j\frac{(v_1 m + v_2 m_2 + 2lT_\omega + r_1 - r_2)^2}{2(v_1 - v_2)}}. \quad (44)$$

The auto-term of $AF_{f, \phi}(\tau, \theta)$ is a straight line across the origin, but the cross-term does not meet this property. Therefore, an effective way to suppress the cross-terms is to carry out low-pass filtering on $AF_f^{T_1}(\tau, \theta)$.

In window T_1 (without losing generality, let $q = 0$), the STWD of the signal f can be represented by its AF

$$W_{f, \phi}^{T_1}(n, k) = 4 \sum_m \sum_n A_f(m, l) \phi'(n, m) e^{-j2(mk + nl)T_\omega}, \quad (45)$$

where $\phi'(n, m) = e^{-\frac{\delta^2}{T_1}(n^2 + m^2)}$, $\delta \geq 2$.

The ϕ' in (45) is equivalent to a low-pass filter, which filters the AF of f , that is, the STWD is equivalent to filtering out the cross-terms in AF. In fact, the STWD in a single window is equivalent to pseudo-WD (PWD) in Cohen class. In other words, the STWD is to add a time domain window on the basis of the time-delay domain window of the PWD. This time domain window can effectively eliminate the cross-terms in the time domain.

The previous research is based on fixed window length and frame length. In order to better adapt to the real-time scene, make the frame length equal to the window length, and then design an optimization algorithm model to adaptively adjust the window length T and the parameter δ in the Gauss window $\phi(n)$.

5.2.2. Optimization model

Rényi entropy can be used to measure the time–frequency energy aggregation. For time–frequency energy density $C(n, k)$, the normalized form of Rényi entropy is

$$H_\alpha(C(n, k)) = \frac{1}{1 - \alpha} \log_2 \sum_n \sum_k \left(\frac{C(n, k)}{\sum_{n'} \sum_{k'} C(n', k')} \right)^\alpha, \quad (46)$$

where $\alpha > 0$ is the order of Rényi entropy. The research shows that when $\alpha = 3$, entropy has a better definition of the time–frequency distribution of most signals, and the stability of entropy measurement is better [18,25]. Therefore, this paper uses the Rényi entropy of the time–frequency distribution with order $\alpha = 3$ as the research object.

Hence, when determining the optimal parameters, the adaptive algorithm is based on the minimum Rényi entropy of the time–frequency distribution. Moreover, the influence of cross-terms in the time domain will be directly reflected in the energy of AF. Assume that for different window lengths $T = 0, 1, \dots, T_{max} - 1$, its corresponding energy value is $E_T = \sum_{m=0}^{T-1} \sum_{l=0}^{T-1} |A_f(m, l)|$, where T_{max} is the maximum window length. Then we calculate the difference of the energy value sequence as $DE_{T'} = E_{T'+1} - E_{T'}$, $T' = 0, 1, \dots, T_{max} - 2$. The optimization model of each window is

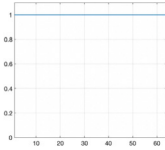
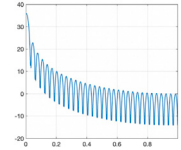
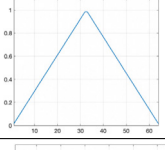
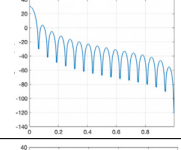
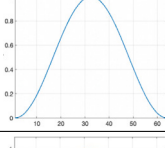
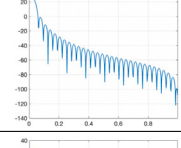
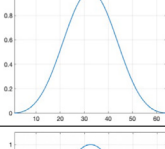
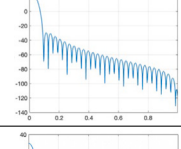
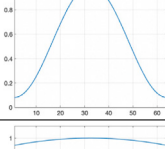
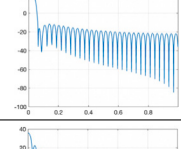
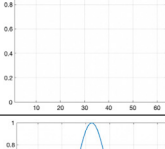
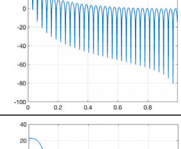
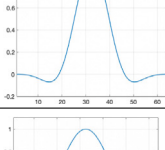
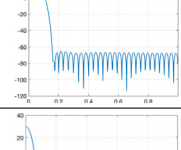
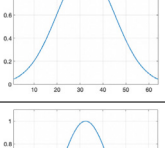
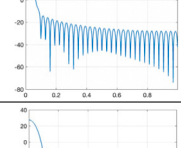
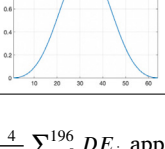
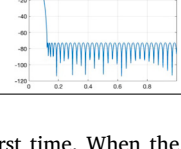
$$\min_{T, \alpha} H_3(A_f(m, l) \phi'(n, m)) \quad (47)$$

$$s.t. \quad DE_{T'} > \frac{4}{T'} \sum_{j=0}^{T'-1} DE_j, \quad T = T' \quad (48)$$

After each window is determined, the end position of the window is used as the starting point of the next window.

In the above optimization model, the constraint condition (48) is mainly used to limit the window length. Because at the junction of signal feature changes, the energy value will produce a change inflection point, so it can be observed through the difference of the energy value. When there is a surge, it means that the cross-term begins to exist. This constraint can improve the overall stability of the model.

Table 1
Different window functions.

Window	Formula	Time domain	Frequency domain
Rectangle	$\phi(n) = 1$		
Triangular	$\phi(n) = \begin{cases} \frac{2n}{N-2}, & n = 0, 1, 2, \dots, \frac{N}{2} - 1 \\ \frac{2N-4-2n}{N-2}, & n = \frac{N}{2}, \dots, N-1 \end{cases}$		
Hanning	$\phi(n) = \frac{1}{2} - \frac{1}{2} \cos\left(\frac{2\pi n}{N}\right)$		
Blackman	$\phi(n) = 0.42 - 0.5 \cos\left(\frac{2\pi n}{N}\right) + 0.08 \cos\left(\frac{4\pi n}{N}\right)$		
Hamming	$\phi(n) = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N}\right)$		
Kaiser	$\phi(n) = \frac{I_0\left[\beta\sqrt{1-\left(1-\frac{2n}{N-1}\right)^2}\right]}{I_0(\beta)}$		
Flat Top	$\phi(n) = 0.21706 - 0.42103 \cos\left(\frac{2\pi n}{N}\right) + 0.28294 \cos\left(\frac{4\pi n}{N}\right) - 0.07897 \cos\left(\frac{6\pi n}{N}\right)$		
Gauss	$\phi(n) = e^{-\frac{1}{2}\left(\frac{6n}{N}\right)^2}, \delta \geq 2$		
Chebyshev	$\phi(n) = \frac{1}{N_{rc}} + \frac{2}{N} \sum_{k=1}^{N/2-1} T_{N/2-1}[\beta \cos(\frac{k\pi}{N})] \cos(\frac{2nk\pi}{N})$		

5.2.3. Simulation

Consider a time-varying signal with varying characteristics

$$f(n) = \begin{cases} e^{j\pi(-3n^2/3920+7n/20)}, & 0 \leq n \leq 196 \\ e^{j\pi(3n^2/4820-1357n/12050)} + e^{j\pi(-27n^2/48200+7247n/48200)}, & 197 \leq n \leq 679 \\ e^{j\pi(-3n^2/3240+583n/540)}, & 680 \leq n \leq 1004 \\ e^{j\pi(-n^2/2215+29403n/44300)} + e^{j\pi(41n^2/44300-9969n/11075)}, & 1005 \leq n \leq 1448 \end{cases} \quad (49)$$

First, the important role of constraint condition (48) is shown in Fig. 7. Set the maximum window length to 500. The first window is determined by Fig. 7(a). It can be seen that when $T'_1 = 197$,

$DE_{197} > \frac{4}{197} \sum_{j=0}^{196} DE_j$ appears for the first time. When the length of the first window is determined to be 197, take the 198th point as the starting point to estimate the length of the second window, as shown in Fig. 7(c). When $T'_2 = 483$, $DE_{483} > \frac{4}{483} \sum_{j=0}^{482} DE_j$ appears for the first time. The estimation of the third window starts from the 681st point, because the sum of the length of the first two windows is 680. Fig. 7(e) shows that when $T'_3 = 325$, $DE_{325} > \frac{4}{325} \sum_{j=0}^{324} DE_j$ appears for the first time. However, there is no T' in Fig. 7(g) to make $DE_{T'} > \frac{4}{T'} \sum_{j=0}^{T'-1} DE_j$ tenable, so it shows that the remaining length is the last window, that is, $T'_4 = N - (T'_1 + T'_2 + T'_3) = 444$. To sum up, this constraint can directly locate the window length. Additionally, we conduct tests to assess the impact of noise. Gaussian white noise with a signal-to-noise ratio (SNR)

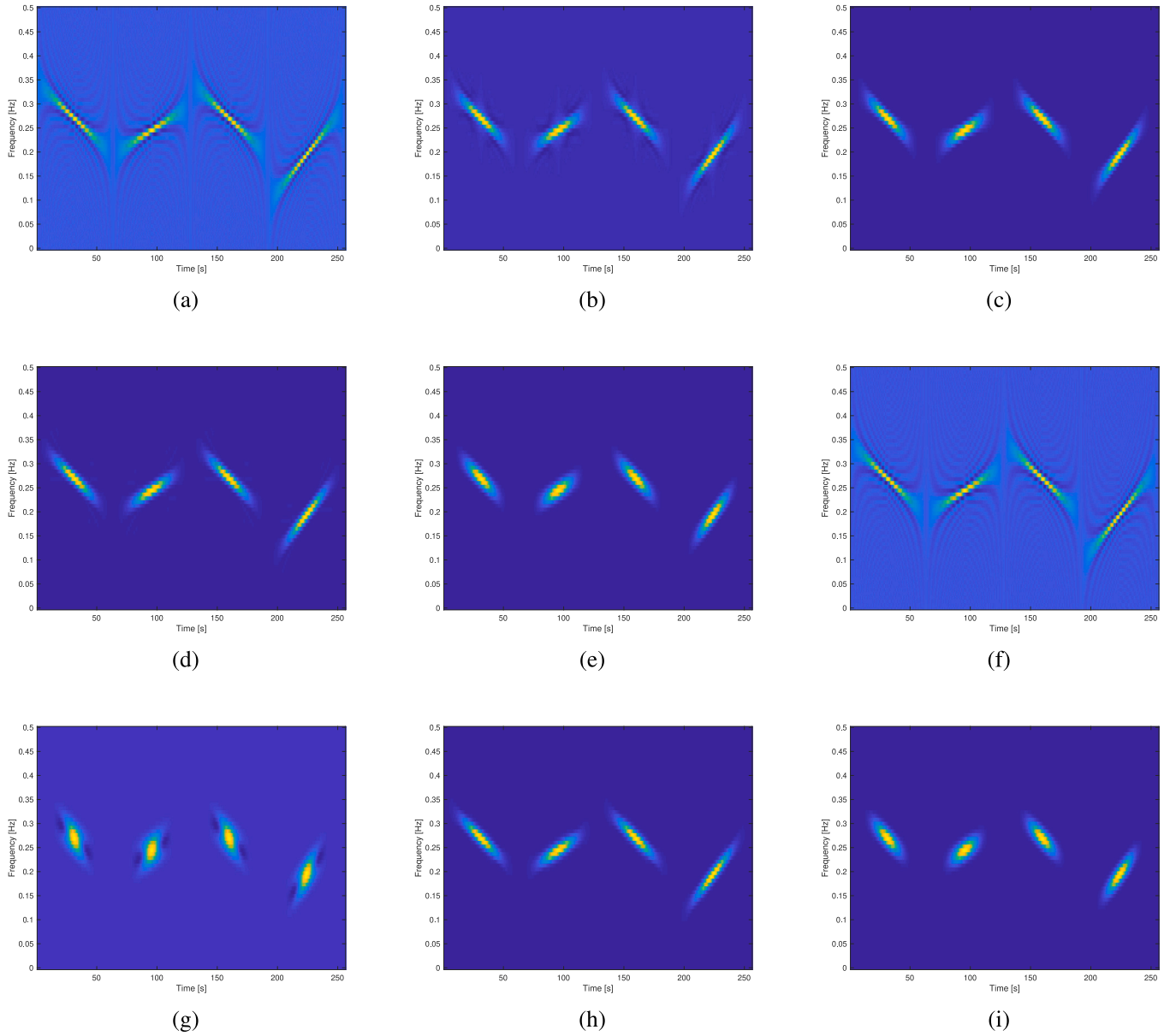


Fig. 5. The DSTWD with different window functions. (a) Rectangle; (b) Triangular; (c) Hanning; (d) Hamming; (e) Blackman; (f) Kaiser; (g) Flat Top; (h) Gauss; (i) Chebyshev.

of 38 dB was added to the signal (49), and the results are depicted in Fig. 7. Even with low levels of noise, we can still achieve ideal results. However, when the noise is high, the accuracy of the model is significantly impacted, necessitating the consideration of denoising methods. Addressing the impact of noise is expected to be a critical focal point for future research endeavors.

Then, take the estimation of the first window as an example, let us observe the change of the objective function value under different window lengths when the parameter $\delta = 2.5$ is fixed. As shown in Fig. 8(a), when the window length is $T = 197$, the objective function value is the minimum. Further, we investigate the change of the objective function under different parameters δ with a fixed window length of $T_1 = 197$, and make a range from 2 to 10. The objective function value is minimum when $\delta_1 = 2$. Finally, we get the estimated result of signal window length as $T_1 = 197, T_2 = 483, T_3 = 325, T_4 = 444$ and $\delta_1 = 2, \delta_2 = 2.48, \delta_3 = 3.79, \delta_4 = 2$.

The resulting time–frequency distribution is depicted in Fig. 9. A comparison with STFT, WD, PWD, SPWD, and the S-method reveals that the ASTWD method proposed in this section provides clearer

representation of signal characteristics while reducing the influence of cross-terms. The ASTWD method's ability to provide a more refined and accurate representation of the time–frequency information enables better understanding and analysis of complex signals. Additionally, the ASTWD method is particularly advantageous for real-time or online signal processing applications. Its adaptive nature allows for dynamic parameter selection, enabling efficient and effective processing of signals in real-time scenarios. This adaptability ensures that the ASTWD method can keep up with the varying characteristics of signals encountered in practical applications. The ASTWD method holds promise for a wide range of applications, where accurate time–frequency analysis is crucial.

6. Conclusion

For time-varying signals with changing characteristics and real-time and online processing of signals, the WD and AF have high limitations due to the existence of cross-items. In this paper, the STWD and STAF that can significantly analyze local change characteristics are proposed.

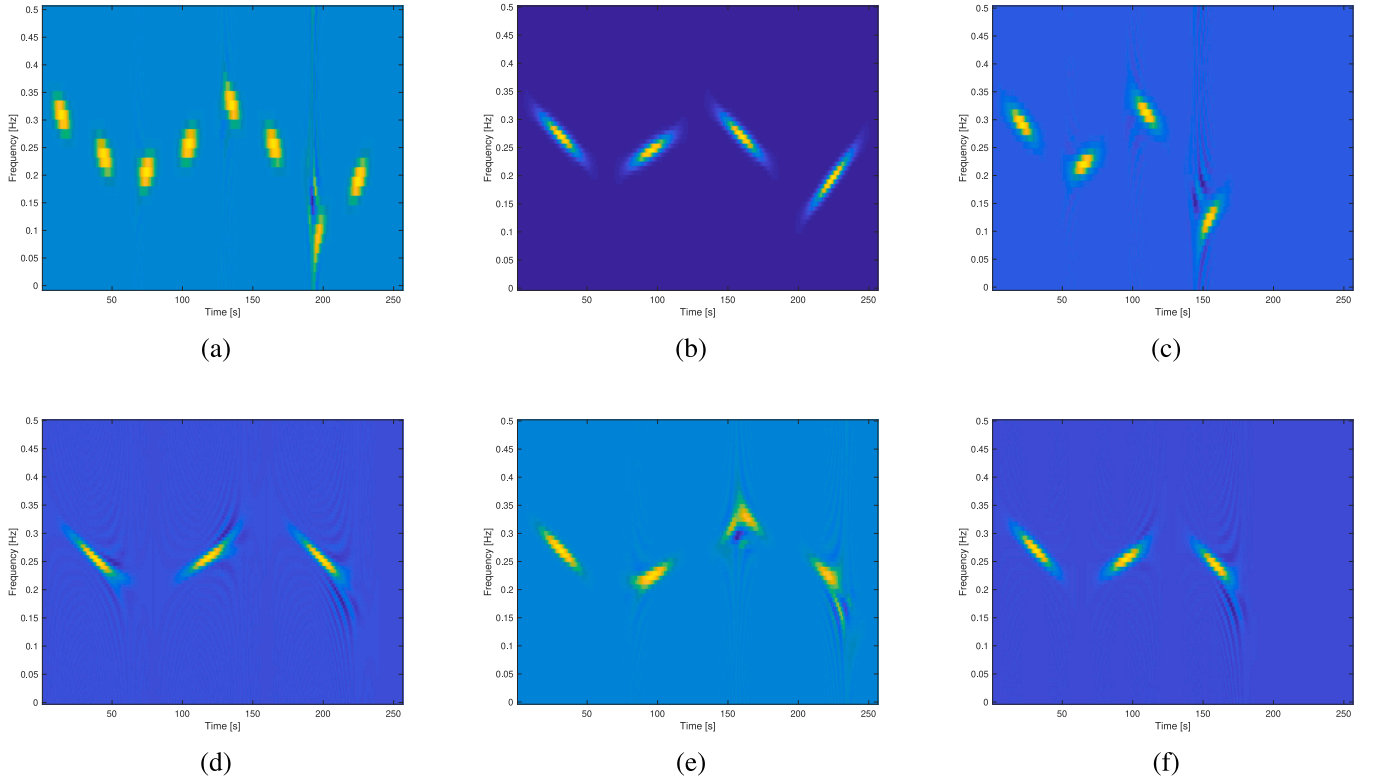


Fig. 6. Influence of different window lengths and frame shifts. (a) $T = 30, S = 30$; (b) $T = 64, S = 64$; (c) $T = 44, S = 60$; (d) $T = 80, S = 64$; (e) $T = 64, S = 50$; (f) $T = 64, S = 70$.

In addition to the basic definitions and properties, we have also derived three types of uncertainty principles. The DSTWD and the process of computerization are also proposed which can be used more widely in practice. Further, the ASTWD based on Rényi entropy is advanced. This method can better achieve real-time and online signal processing. However, practical applications often involve additional challenges, such as signal randomness, signal denoising, and parameter estimation and detection. These aspects represent key research questions that need to be addressed in future studies.

CRediT authorship contribution statement

Jian Yi Chen: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Bing Zhao Li:** Formal analysis, Funding acquisition, Project administration, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A. Proof of Heisenberg's uncertainty principle

Proof. Based on (22), the inverse transform and the Parseval formula for the FT, we have

$$\int_{\mathbb{R}} (t - t^0)^2 \left| F^{-1}[F(u)](t) \right|^2 dt \int_{\mathbb{R}} (u - u^0)^2 |F(u)|^2 du \geq \frac{1}{16\pi^2} \left(\int_{\mathbb{R}} |F(u)|^2 du \right)^2. \quad (\text{A.1})$$

Since $W_{f,\phi}(t, u) \in L^2(\mathbb{R})$, we can replace the FT by the STWD of f in (A.1), we get

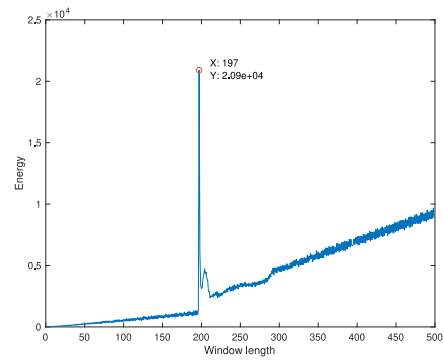
$$\begin{aligned} & \int_{\mathbb{R}} (t - t^0)^2 \left| f(t + \frac{\tau}{2}) \phi(t - p + \frac{\tau}{2}) f^*(t - \frac{\tau}{2}) \phi^*(t - p - \frac{\tau}{2}) \right|^2 dt \\ & \int_{\mathbb{R}} (u - u^0)^2 |W_{f,\phi}(t, u)|^2 du \\ & \geq \frac{1}{16\pi^2} \left(\int_{\mathbb{R}} |W_{f,\phi}(t, u)|^2 du \right)^2, \end{aligned} \quad (\text{A.2})$$

taking the square root on both sides of (A.2) and integrating both sides with respect to dt , and then applying the Cauchy-Schwarz inequality and the Parseval formula in (21), we obtain

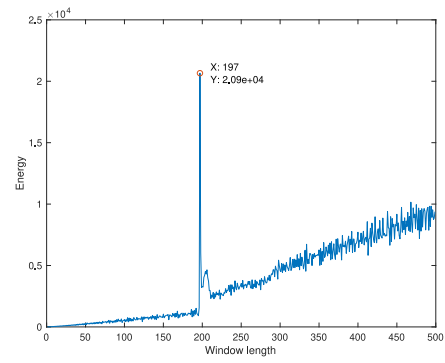
$$\begin{aligned} & \left(\iint_{\mathbb{R}^2} (t - t^0)^2 \left| f(t + \frac{\tau}{2}) \phi(t - p + \frac{\tau}{2}) f^*(t - \frac{\tau}{2}) \phi^*(t - p - \frac{\tau}{2}) \right|^2 d\tau dt \right)^{\frac{1}{2}} \\ & \times \left(\iint_{\mathbb{R}^2} (u - u^0)^2 |W_{f,\phi}(t, u)|^2 du d\tau \right)^{\frac{1}{2}} \geq \frac{1}{4\pi} \|f\|_2^4 \|\phi\|_2^4, \end{aligned} \quad (\text{A.3})$$

where the first term on the left side can be deformed into

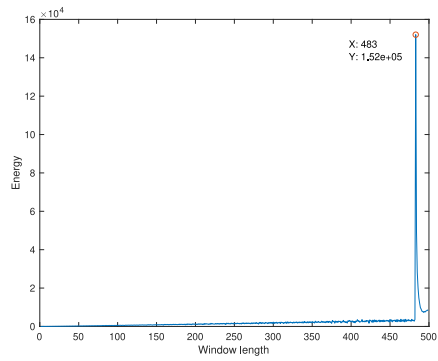
$$\begin{aligned} & \iint_{\mathbb{R}^2} (t - t^0)^2 \left| f(t + \frac{\tau}{2}) \phi(t - p + \frac{\tau}{2}) f^*(t - \frac{\tau}{2}) \phi^*(t - p - \frac{\tau}{2}) \right|^2 d\tau dt \\ & = \int_{\mathbb{R}} (t - t^0)^2 |f(t)|^2 dt \int_{\mathbb{R}} |\phi(t - p)|^2 dt \int_{\mathbb{R}} (t - t^0)^2 |f(t - \tau)|^2 d\tau \\ & \times \int_{\mathbb{R}} (t - t^0)^2 |\phi(t - p - \tau)|^2 d\tau = \int_{\mathbb{R}} (t - t^0)^2 |f(t)|^2 dt \|\phi\|_2^4 \|f\|_2^2. \end{aligned} \quad (\text{A.4})$$



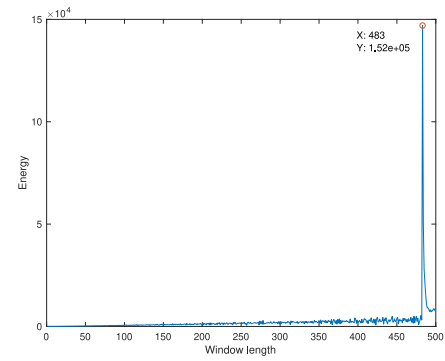
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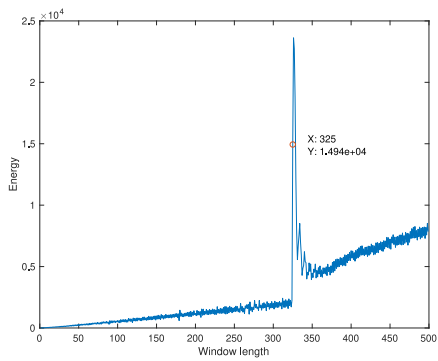
(b)



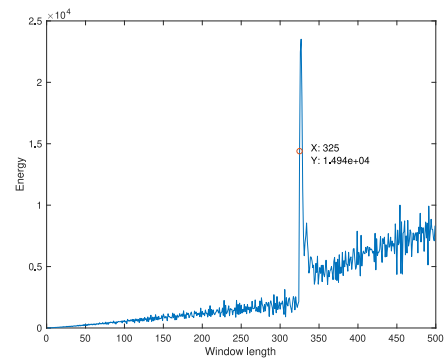
(c)



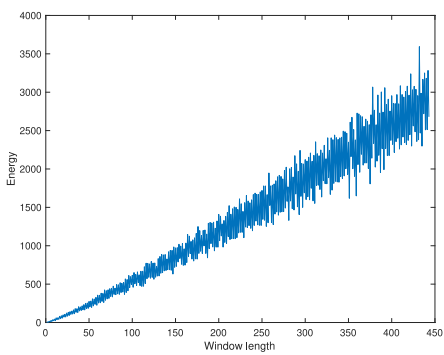
(d)



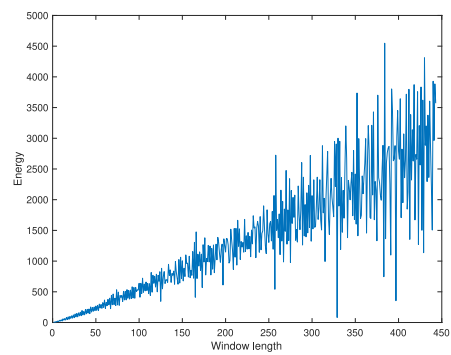
(e)



(f)



(g)



(h)

Fig. 7. Differential sequence. (a) First window. (b) First window with noise. (c) Second window. (d) Second window with noise. (e) Third window. (f) Third window with noise. (g) Last window. (h) Last window with noise.

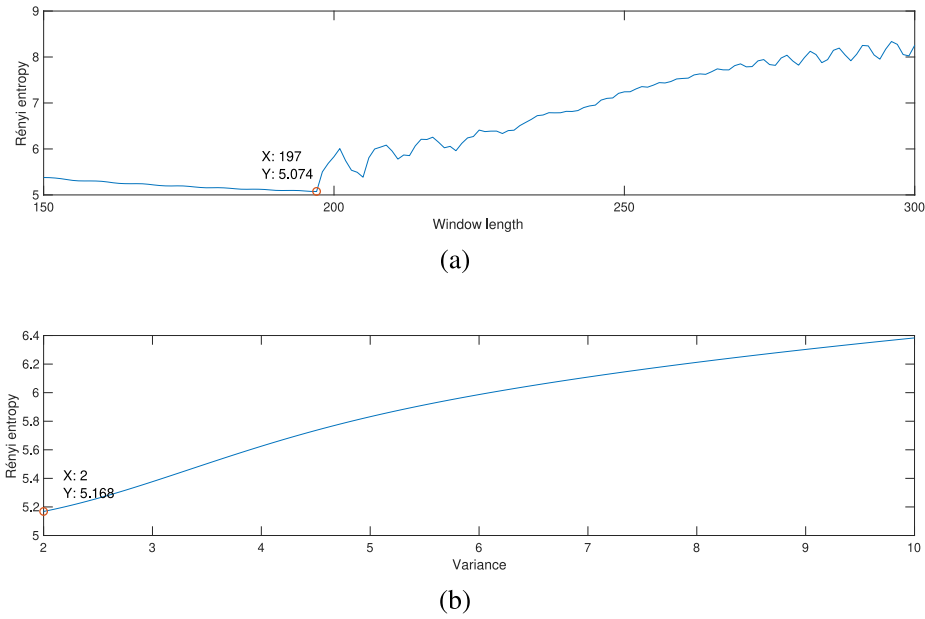


Fig. 8. (a) Different window length T with fixed α . (b) Different α with fixed window length T .

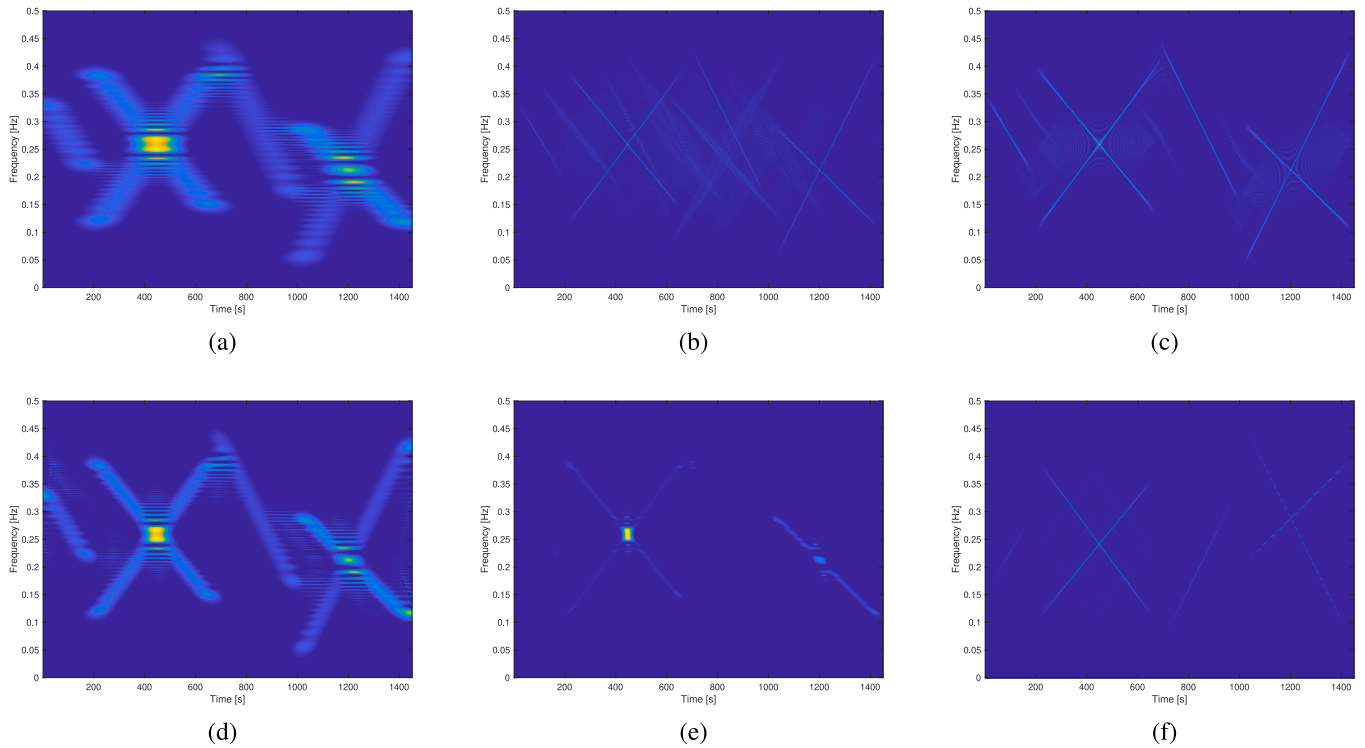


Fig. 9. (a) The STFT on the time-varying signal f with varying characteristics in (39). (b) The WD on f . (c) The PWD on f . (d) The SPWD on f . (e) The S-method on f . (f) The ASTWD on f .

Therefore, taking (23) and (A.4) into (A.3), we have

$$\geq \frac{1}{4\pi} \|f\|_2^4 \|\phi\|_2^4. \quad (\text{A.5})$$

$$\left(\int_{\mathbb{R}} (t - t_\phi^0)^2 |f(t)|^2 dt \|\phi\|_2^4 \|f\|_2^2 \right)^{\frac{1}{2}} \left(\iint_{\mathbb{R}^2} (u - u_\phi^0)^2 |W_{f,\phi}(t, u)|^2 du dt \right)^{\frac{1}{2}}$$

Dividing both sides of the above inequality by $\|\phi\|_2^2 \|f\|_2$, we obtain the desired result. $\square$

Appendix B. Proof of Logarithmic uncertainty principle

Proof. Based on (25), the inverse transform and the Parseval formula for the FT, we get

$$\int_{\mathbb{R}} \left| F^{-1}[F(u)](t) \right|^2 \ln |t| dt + \int_{\mathbb{R}} |F(u)|^2 \ln |u| du \geq D \int_{\mathbb{R}} |F(u)|^2 du. \quad (\text{B.1})$$

Replacing the FT by the STWD of f in (B.1)

$$\begin{aligned} & \int_{\mathbb{R}} \left| f\left(t + \frac{\tau}{2}\right) \phi\left(t - p + \frac{\tau}{2}\right) f^*\left(t - \frac{\tau}{2}\right) \phi^*\left(t - p - \frac{\tau}{2}\right) \right|^2 \ln |t| d\tau \\ & + \int_{\mathbb{R}} \left| W_{f,\phi}(t, u) \right|^2 \ln |u| du \\ & \geq D \int_{\mathbb{R}} \left| W_{f,\phi}(t, u) \right|^2 du. \end{aligned} \quad (\text{B.2})$$

Then using the Parseval formula in (21) and integrating both sides with respect to dt , we have

$$\begin{aligned} & \iint_{\mathbb{R}^2} \left| f\left(t + \frac{\tau}{2}\right) \phi\left(t - p + \frac{\tau}{2}\right) f^*\left(t - \frac{\tau}{2}\right) \phi^*\left(t - p - \frac{\tau}{2}\right) \right|^2 \ln |t| d\tau dt \\ & + \iint_{\mathbb{R}^2} \left| W_{f,\phi}(t, u) \right|^2 \ln |u| du dt \geq D \|f\|^4 \|\phi\|^4, \end{aligned} \quad (\text{B.3})$$

where the first term on the left side can be deformed into

$$\begin{aligned} & \iint_{\mathbb{R}^2} \left| f\left(t + \frac{\tau}{2}\right) \phi\left(t - p + \frac{\tau}{2}\right) f^*\left(t - \frac{\tau}{2}\right) \phi^*\left(t - p - \frac{\tau}{2}\right) \right|^2 \ln |t| d\tau dt \\ & = \|\phi\|^4 \|f\|^2 \int_{\mathbb{R}} \ln |t| |f(t)|^2 dt. \end{aligned} \quad (\text{B.4})$$

Taking (B.4) into (B.3), we obtain

$$\|\phi\|^4 \|f\|^2 \int_{\mathbb{R}} \ln |t| |f(t)|^2 dt + \iint_{\mathbb{R}^2} \left| W_{f,\phi}(t, u) \right|^2 \ln |u| du dt \geq D \|f\|^4 \|\phi\|^4. \quad (\text{B.5})$$

The proof is completed. $\square$

Appendix C. Proof of Lieb's uncertainty principle

C.1. Weak uncertainty principle

Proof. Based on

$$\begin{aligned} W_{f,\phi}(t, u) &= \int_{\mathbb{R}} (f_{\phi}^{\tau/2} \otimes f_{\phi}^{\tau/2})(t) e^{-j\tau u} d\tau \\ &= \left\langle f\left(t + \frac{\tau}{2}\right) \phi\left(t - p + \frac{\tau}{2}\right), f^*\left(t - \frac{\tau}{2}\right) \phi^*\left(t - p - \frac{\tau}{2}\right) e^{-j\tau u} \right\rangle, \end{aligned}$$

where $\langle \cdot \rangle$ represents the inner product, and then applying the Cauchy-Schwarz inequality

$$\begin{aligned} \left| W_{f,\phi}(t, u) \right|^2 &= \left| \left\langle f\left(t + \frac{\tau}{2}\right) \phi\left(t - p + \frac{\tau}{2}\right), f^*\left(t - \frac{\tau}{2}\right) \phi^*\left(t - p - \frac{\tau}{2}\right) e^{-j\tau u} \right\rangle \right|^2 \\ &\leq \|f\|_2^2 \|\phi\|_2^2 = 1. \end{aligned}$$

Therefore, we obtain

$$1 - \varepsilon \leq \iint_M \left| W_{f,\phi}(t, u) \right|^2 dt du \leq \|W_{f,\phi}\|_{\infty}^2 |M| \leq |M|. \quad \square \quad (\text{C.1})$$

C.2. Lieb's uncertainty principle

Proof. Since the STWD is the FT of $(f_{\phi}^{\tau/2} \otimes f_{\phi}^{\tau/2})(t)$, then using Lemma 1, we get

$$\begin{aligned} \int_{\mathbb{R}} \left| W_{f,\phi}(t, u) \right|^p du &\leq C_q^p \left[\int_{\mathbb{R}} \left| f\left(t + \frac{\tau}{2}\right) \phi\left(t - p + \frac{\tau}{2}\right) \right|^q \left| f^*\left(t - \frac{\tau}{2}\right) \phi^*\left(t - p - \frac{\tau}{2}\right) \right|^q \right]^{\frac{p}{q}} \\ &= C_q^p \left[(|f\phi|^q * |f^{-}\phi^{-}|^q)(\tau) \right]^{\frac{p}{q}}, \end{aligned} \quad (\text{C.2})$$

where $f^{-}(x) = f^{*}(-x)$, and $*$ denotes the convolution operator. Next, let $r = p/q > 1$, $m = 2/q \geq 1$, $n = 2/q \geq 1$, and applying Lemma 2, for $J = [|f\phi|^q * |f^{-}\phi^{-}|^q](\tau)$, we have

$$\begin{aligned} \iint_M \left| W_{f,\phi}(t, u) \right|^2 dt du &\leq C_q^p \|J\|_r' \geq C_q^p \left(\frac{C_m C_n}{C_r} \right)^r \left[\|f\phi\|_m \|f^{-}\phi^{-}\|_n \right]^r \\ &= C_q^p \left(\frac{C_m C_n}{C_r} \right)^r \left[\|f\phi\|_2 \|f\phi\|_2 \right]^p = \frac{2}{p} \left[\|f\phi\|_2^2 \right]^p \\ &\leq \frac{2}{p} \left[\left(\|f\|_2^{1/2} \|\phi\|_2^{1/2} \right)^2 \right]^p = \frac{2}{p} \left[\|f\|_2^2 \|\phi\|_2^2 \right]^p. \quad \square \end{aligned}$$

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